

1. Workload

2. ISA & ABI

3. Compute Org

4. Memory & NoC

5. Power Intent

6. Compiler/RT

7. Physical Node

8. Verification



CHAPTER STACK SCOPE — ACTIVE LAYERS 2-6 (ABI TO NETLIST)

Methods: Generation, Prediction & Optimization

1. Generation

- LLM Code Completion
- Pattern Synthesis Engine
- Microarch Proposer
- Parameter Sweeper

2. Prediction

- GNN Performance Surrogate
- Early STA Slack Estimator
- Thermal Map Predictor
- Confidence Interval Filter

3. Optimization

- Reinforcement Learning DSE
- Bayesian Parameter Search
- Evolutionary Microarch
- Pareto Selection Engine

4. Method Signoff

- Search Efficiency Check
- Surrogate Accuracy Verification
- Method vs Baseline Claim
- Execution Budget Signoff

Surrogate Retraining & Optimizer Tuning Loop